

1 FEASIBILITY STUDY

A feasibility study is conducted to determine whether the proposed NeuroSync system is practical, achievable, and beneficial before development begins. It evaluates the project from economic, technical, and behavioural perspectives to ensure that the system can be successfully implemented and accepted by its intended users.

1.1 Economic Feasibility

Economic feasibility evaluates whether the benefits of the project justify the development and operational costs. NeuroSync is designed using cost-effective and open-source technologies, which significantly reduce the overall implementation cost.

The project utilizes technologies such as React.js for the frontend, Node.js with Express.js for the backend, MongoDB as the database, and Python for AI-based emotion and sentiment analysis. These technologies are open source and freely available. During the development phase, the project can be hosted locally, and cloud deployment can be carried out later using affordable platforms. The expected costs include domain registration, cloud hosting, API usage (if advanced AI services are integrated), and maintenance.

The platform provides long-term value by helping users improve mental well-being, productivity, emotional awareness, and healthy habits. Therefore, the overall benefits outweigh the implementation cost, making the project economically feasible.

1.2 Technical Feasibility

Technical feasibility determines whether the required technology, tools, and technical expertise are available to develop the system successfully.

NeuroSync can be implemented using modern web technologies such as React.js, Node.js, Express.js, MongoDB, Python, HTML, CSS, and JavaScript. AI functionalities such as sentiment analysis, mood prediction, chatbot integration, and recommendation systems can be developed using Python libraries and machine learning models. Authentication can be implemented using JWT or Firebase Authentication, while cloud storage and deployment can be managed using platforms such as AWS, Firebase, or Vercel.

The required software tools include Visual Studio Code, Git, Node.js, MongoDB, Python, and modern web browsers. Since these technologies are widely available and well-supported, the project is technically feasible.

1.3 Behavioral Feasibility

Behavioral feasibility examines whether users will accept and effectively use the proposed system.

NeuroSync is designed with a simple, user-friendly, and responsive interface that can be easily used by students, working professionals, and individuals seeking to improve their mental wellness and productivity. The platform provides personalized recommendations, AI-based emotional support, mood tracking, productivity insights, and wellness guidance, making it engaging and beneficial for users.

The system prioritizes user privacy and data security by implementing secure authentication and protecting personal information. Since the platform addresses real-life challenges such as stress management, emotional well-being, and productivity enhancement, users are expected to adopt the system with minimal training. Hence, the project is behaviourally feasible.